



UNIVERSITY INTERSCHOLASTIC LEAGUE

Science

Invitational B • 2022



GENERAL DIRECTIONS:

- **DO NOT OPEN EXAM UNTIL TOLD TO DO SO.**
- Contestants may take up to two hours to complete the contest. If you are in the process of actually writing an answer when the signal to stop is given, you may finish writing that answer.
- Papers may not be turned in until 30 minutes have elapsed. If you finish the test in less than 30 minutes, remain at your seat and retain your paper until told to do otherwise. You may use this time to check your answers.
- All answers must be written on the answer sheet provided. Indicate your answers in the appropriate blanks provided on the answer sheet. Write clearly and legibly!
- You may place as many notations as you desire anywhere on the test paper but not on the answer sheet, which is reserved for answers only.
- You may use additional scratch paper provided by the contest director.
- All questions have ONE and only ONE correct (BEST) answer. There is a penalty for all incorrect answers.
- If a question is omitted, no points are given or subtracted.
- The back two pages of this test include a copy of the periodic table of the elements, as well as listings of other scientific relationships. You may use this information during the contest and may detach the back page from the test if you wish.
- A simple scientific calculator is sufficient for the high school Science contest. **The UIL provides a list of approved calculators that meet the criteria for use in the Science contest. No other calculators are permitted during the contest.** The Science Contest Approved Calculator List is available in the current Science Contest Handbook and on the UIL website. Contest directors will perform a brief visual inspection to confirm that all contestants are using only approved calculators. Each contestant may use up to two approved calculators during the contest.

- B01. The nucleus of a eukaryotic cell is surrounded by a double membrane called the
- A) rough endoplasmic reticulum.
 - B) Golgi apparatus.
 - C) nuclear envelope.
 - D) nuclear pores.
 - E) cytoplasmic membrane.

- B02. In Mendelian classical genetics, what percent of the offspring from the following genetic cross would phenotypically express both dominant traits?

AAbb x AaBb

- A) 0%
- B) 25%
- C) 50%
- D) 75%
- E) 100%

- B03. Which of the following statements about DNA replication is incorrect?

- A) The replication of DNA produces RNA transcripts.
- B) DNA polymerases replicate DNA.
- C) Replication is bidirectional.
- D) DNA replication must occur prior to cell division.
- E) DNA replication occurs within the nucleus of eukaryotic cells and the nucleoid region of prokaryotic cells.

- B04. The joining together of DNA pieces from different sources is called

- A) Polymerase Chain Reaction.
- B) recombinant DNA technology.
- C) DNA fingerprinting.
- D) whole organism cloning.
- E) Northern blotting.

- B05. Astrocytes, ependymal, microglia, oligodendrocytes, and Schwann cells are all accessory cells within
- A) epithelial tissue.
 - B) connective tissue.
 - C) muscle tissue.
 - D) nervous tissue.

- B06. In the Fluid Mosaic Model of biological membranes, the two leaflets of the membrane are asymmetrical in both structure and function. Knowing that membranes are asymmetrical, what could you hypothesize about them?

- A) Phospholipids can congregate on one side of the membrane and distort the structure.
- B) Glycoproteins, such as those found on red blood cells, are likely embedded within the outer leaflet of the membrane.
- C) Enzymes are responsible for moving phospholipids from one leaflet to another, rather than the movement occurring spontaneously.
- D) Membrane asymmetry sequesters membrane-involved physiological responses and cellular activities to specific locations on the membrane.
- E) All of the above are potential hypotheses regarding asymmetrical membranes.

B07. Which of the following is not responsible in the

- B07. Examine the microscopic image of the epithelial tissue lining a kidney tubule. This image is a cross-section of a collecting duct. The structure identified by the arrow is the



http://medcell.med.yale.edu/histology/urinary_system_lab/collecting_ducts.php

- A) cell wall.
- B) connective tissue.
- C) basement membrane.
- D) plasma membrane.
- E) nucleus.

- B08. In a chemical reaction catalyzed by an enzyme, the _____ binds to the _____ site.
- A) product; substrate
 - B) substrate; active
 - C) product; allosteric
 - D) product; active
 - E) substrate; allosteric
- B09. In a population at Hardy-Weinberg equilibrium, 300 out of 1000 individuals in the population have a recessive phenotype. Which Hardy-Weinberg variable can be solved for first?
- A) p^2
 - B) p
 - C) q^2
 - D) q
 - E) $2pq$
- B10. The ribosomes on the rough endoplasmic reticulum in eukaryotic cells
- A) synthesize cytosolic proteins.
 - B) generate enzymes only.
 - C) make lipids.
 - D) make proteins designated for secretion or export.
 - E) are permanently attached to the organelle.
- B11. Regulation of the cell cycle involves the expression of cyclins that bind to cyclin-dependent kinases to effect an outcome. In which cell cycle phase do you think M cyclins are expressed at their highest level?
- A) Mitosis
 - B) G1/S phase
 - C) G1 phase
 - D) S phase
 - E) G0 phase
- B12. What would likely happen if a mutation occurred in the 3'-untranslated region of a gene?
- A) Regulation of gene expression would be affected.
 - B) The stop codon would not be read correctly.
 - C) Nothing, as this area is downstream of the gene and not translated into protein.
 - D) An amino acid substitution would occur.
 - E) The start codon would not be identified, and the protein would not be made.
- B13. Which of the following eukaryotic pathogens is vector-borne?
- A) *Ancylostoma duodenale*
 - B) *Plasmodium falciparum*
 - C) *Naegleria fowleri*
 - D) *Entamoeba histolytica*
 - E) *Aspergillus niger*
- B14. Examine the following answer choices. Which biological organization level would represent the broadest category relative to the other answer choices?
- A) atoms
 - B) ecosystem
 - C) community
 - D) organs
 - E) molecules
- B15. Which of the following is not responsible in the formation of acid rain?
- A) deforestation
 - B) burning of fossil fuels
 - C) nitrogen oxides and sulfur dioxides
 - D) volcanic activity
 - E) manufacturing facilities
- B16. A gene in a population that confers a selective advantage would
- A) only affect the individual in the population and not the population as a whole in terms of evolution.
 - B) likely decrease the reproductive success.
 - C) create a bottleneck effect.
 - D) increase in frequency over time.
 - E) none of the above would occur.
- B17. All of the following are examples of convergent evolution except
- A) the body structure of sharks and dolphins.
 - B) the presence of wings on insects and bats.
 - C) the eyes of vertebrates and cephalopods.
 - D) analogous structures.
 - E) mammalian forearm structure.

B18. In temperate climates, the age of a tree can be determined by counting its annual growth rings. Which structure is actually being counted?

- A) bark
- B) phloem
- C) epidermis
- D) xylem
- E) apical meristem

B19. The Centers for Disease Control and Prevention has identified organic baby spinach as the source of an *E. coli* outbreak in nine states. Which of the following is not associated with this particular outbreak or strain?

- A) *E. coli* O157:H7
- B) hemolytic uremic syndrome
- C) bloody diarrhea
- D) Shiga-toxin producing *E. coli*
- E) ESBL-*E. coli*

B20. The reactions occurring within the _____ produce a majority of the carbon dioxide in eukaryotic cellular respiration.

- A) mitochondria.
- B) chloroplast.
- C) nucleus.
- D) cytosol.
- E) plasma membrane.

C06. Containers A and B have the same diameter and are at the same temperature. If the pressure in container A is 1.50 atm, what is the pressure in container B?



- A) 1.500 atm
- B) 1.375 atm
- C) 1.250 atm
- D) 1.125 atm
- E) 1.000 atm

C07. How many grams of solid PbI₂ would be produced by adding excess KI to 88 mL of a 0.052 M Pb(NO₃)₂ solution?

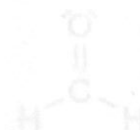
- A) 0.91 g
- B) 1.5 g
- C) 2.1 g
- D) 3.0 g
- E) 3.7 g

C08. An atom of Element A has an emission spectrum that includes a red line at 690 nm. The emission spectrum of an atom of Element B also has a red line, but at 632 nm. What can you say about the red photons coming off these two atoms?

- A) The red photon from Element A is traveling faster than the red photon from Element B.
- B) The red photon from Element A is larger than the red photon from Element B.
- C) The red photon from Element A is higher energy than the red photon from Element B.
- D) The red photon from Element A is lower energy than the red photon from Element B.
- E) The two photons are equal in energy, because they are both in the red region of the visible spectrum.

C09. What are the strongest intermolecular forces present in liquid hexamethylenediamine, shown here?

- A) Hydrogen bonding
- B) Covalent bonds
- C) Dipole-dipole forces
- D) Dispersion forces
- E) Liquid bonds



C10. In which one of the following compounds is the metal in a +3 oxidation state?

- A) K₂O
- B) FeCl₂
- C) SnCl₄
- D) V₂O₅
- E) Al₂O₃

C01. Which of these elements can be found in the 4d section of the periodic table?

- A) Ag
- B) Cr
- C) K
- D) La
- E) Te

C02. What is the mass of half a mole of sodium bromite (NaBrO_2)?

- A) 59.45 g
- B) 67.45 g
- C) 118.89 g
- D) 134.89 g
- E) 269.78 g

C03. How many grams of solid PbI_2 would be produced by adding excess KI to 88 mL of a 0.052 M $\text{Pb}(\text{NO}_3)_2$ solution?

- A) 0.95 g
- B) 1.5 g
- C) 2.1 g
- D) 2.5 g
- E) 3.7 g

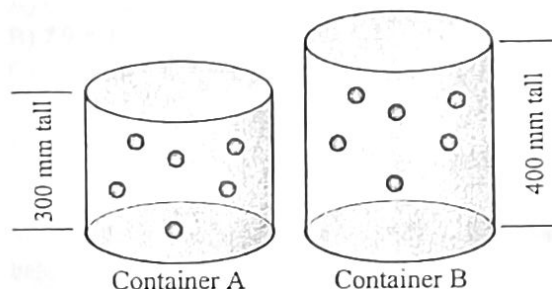
C04. An atom of Element A has an emission spectrum that includes a red line at 695 nm. The emission spectrum of an atom of Element B also has a red line, but at 632 nm. What can you say about the red photons coming off these two atoms?

- A) The red photon from Element A is traveling faster than the red photon from Element B.
- B) The red photon from Element A is larger than the red photon from Element B.
- C) The red photon from Element A is higher energy than the red photon from Element B.
- D) The red photon from Element A is lower energy than the red photon from Element B.
- E) The two photons are equal in energy because they are both in the red region of the visible spectrum.

C05. What is the shape of a phosphorus trichloride molecule?

- A) trigonal pyramidal
- B) trigonal planar
- C) linear
- D) tetrahedral
- E) square planar

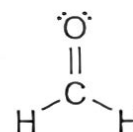
C06. Containers A and B have the same diameter and are at the same temperature. If the pressure in container A is 1.50 atm, what is the pressure in container B?



- A) 1.500 atm
- B) 1.375 atm
- C) 1.250 atm
- D) 1.125 atm
- E) 1.000 atm

C07. What are the strongest intermolecular forces present in liquid formaldehyde, shown here?

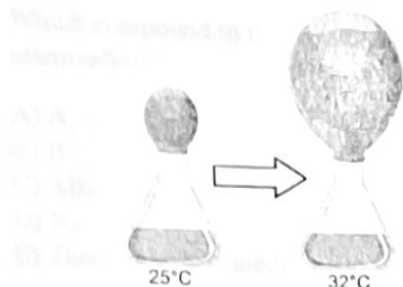
- A) Hydrogen bonding
- B) Covalent bonds
- C) Dipole-dipole forces
- D) Dispersion forces
- E) Liquid bonds



C08. In which one of these compounds is the metal in a +3 oxidation state?

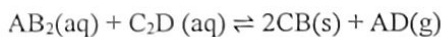
- A) K_2O
- B) CaO
- C) SnO_2
- D) V_2O_5
- E) Al_2O_3

- C09. A chemical reaction between two compounds dissolved in water is carried out in a flask covered with a balloon and the following changes are observed. Is this reaction endothermic or exothermic, and how do you know?



- A) Exothermic because the volume increased.
 B) Exothermic because the solution temperature increased.
 C) Endothermic because the volume increased.
 D) Endothermic because the solution temperature increased.
 E) Two of the above answers are correct.
- C10. How much energy would it take to melt a pound of ice (454 grams) at 0°C and heat the water to 25°C?
- A) 199 kJ
 B) 152 kJ
 C) 47.5 kJ
 D) 1030 kJ
 E) 1070 kJ

- C11. Which of these is the correct equilibrium expression for the reaction



- A) $K_{\text{eq}} = \frac{[\text{AB}_2][\text{C}_2\text{D}]}{[\text{CB}]^2[\text{AD}]}$
 B) $K_{\text{eq}} = \frac{[\text{AB}_2][\text{C}_2\text{D}]}{[\text{AD}]}$
 C) $K_{\text{eq}} = \frac{[2\text{CB}][\text{AD}]}{[\text{AB}_2][\text{C}_2\text{D}]}$
 D) $K_{\text{eq}} = \frac{[\text{CB}]^2[\text{AD}]}{[\text{AB}_2][\text{C}_2\text{D}]}$
 E) $K_{\text{eq}} = \frac{[\text{AD}]}{[\text{AB}_2][\text{C}_2\text{D}]}$

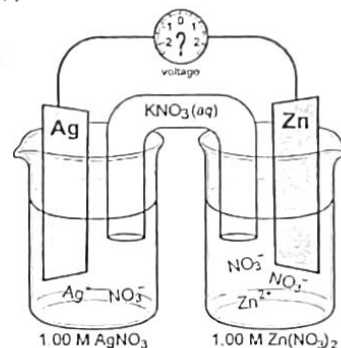
- C12. Which of these compounds would result in a solution with the highest pH if 0.03 mole of the compound was dissolved in 1.0 kg of water?

- A) KOH
 B) KCl
 C) H_2SO_4
 D) BaCl_2
 E) $\text{Ba}(\text{OH})_2$

- C13. If insoluble salt XY has a K_{sp} of 6.2×10^{-5} , how many moles of XY will dissolve in 1 L of solution?

- A) 6.2×10^{-5}
 B) 7.9×10^{-3}
 C) 2.0×10^{-3}
 D) 2.5×10^{-4}
 E) 3.8×10^{-9}

- C14. What is the potential of the electrolytic cell shown below?

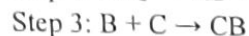
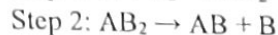


- A) +1.56 V
 B) -1.56 V
 C) +0.04 V
 D) -0.04 V
 E) +0.80 V

- C15. Based on periodic trends, which of these ions would be expected to have the largest radius?

- A) Cl^-
 B) Ga^{3+}
 C) K^+
 D) Ca^{2+}
 E) F^-

- C16. The overall reaction $A + B_2 + C \rightarrow AB + CB$ has the following reaction mechanism:



Which compound in the mechanism is an intermediate?

- A) A
- B) B
- C) AB_2
- D) B_2
- E) There is no intermediate.

- C17. A tank of compressed helium gas holds 12.8 moles of helium gas. How many balloons can you fill from one tank of compressed helium gas if at 1.00 atm pressure and 25.0°C , each balloon has a volume of 6.25 L?

- A) 40
- B) 50
- C) 60
- D) 65
- E) 75

- C18. What is the mass of 8.50×10^{25} atoms of iron?

- A) 7.89 kg
- B) 141 kg
- C) 39.6 kg
- D) 2.21 kg
- E) 2.53 kg

- C19. When salt is dissolved in water, what happens to the freezing point and boiling point of the solution compared to those of pure water?

- A) They both increase.
- B) They both decrease.
- C) The boiling point goes up and the freezing point goes down.
- D) The boiling point goes down and the freezing point goes up.
- E) They do not change.

- C20. If you dissolve 100. grams of solid CaCl_2 in water and then dilute it to 500. mL, what will the concentration of chloride ions be in the final solution?

- A) 1.80 M
- B) 3.60 M
- C) 5.30 M
- D) 0.90 M
- E) 2.65 M

- P07. You are pushing a 50.0 kg crate of equipment up a frictionless inclined plane. The plane is inclined at 22.0° with respect to the horizontal, and the 200 N force you apply to the crate is parallel to the plane (as shown). What is the velocity of the crate after 2.50 seconds?



- A) 0.822 m/s
- B) 4.30 m/s
- C) 9.18 m/s
- D) 10.0 m/s
- E) 14.5 m/s

- P08. A small 150 kg cart rolls frictionlessly across a horizontal table at a speed of 1.00 m/s (to the right). The cart collides with a larger 2.50 kg cart that is stationary. After the collision, the small cart bounces directly backwards (to the left) with a speed of 1.30 m/s. What is the speed of the large cart after the collision?

- A) 0.852 m/s
- B) 1.02 m/s
- C) 1.19 m/s
- D) 1.83 m/s
- E) 2.57 m/s

- P01. According to Kaku, the concept of a field is one of the most important concepts in all of physics. Who invented the concept of a field?

A) Galileo
B) Newton
C) Faraday
D) Maxwell
E) Hertz

- P02. According to Kaku, the first experimental test of Einstein's theory of General Relativity was ____.

A) the bending of light around the Sun.
B) the erratic orbit of Mercury.
C) the conversion of mass into energy.
D) the discovery of Neptune.
E) the speeding up of time when in orbit.

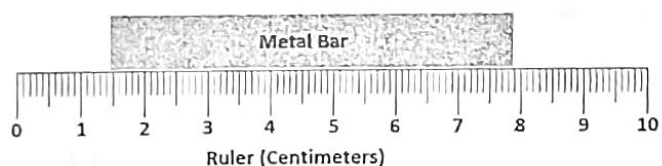
- P03. According to Kaku, electrons are both particles and waves. What is it about an electron that is a wave, described by a wave function?

A) the magnetic moment of the electron
B) the momentum of the electron
C) the electric charge of the electron
D) the probability of finding the electron
E) the trajectory of the electron

- P04. Long period comets are thought to originate from a spherical shell of icy objects orbiting thousands of Astronomical Units from the Sun. What is this spherical shell of objects called?

A) The Kuiper Belt
B) The Hill Cloud
C) The Asteroid Belt
D) The Scattered Disk
E) The Oort Cloud

- P05. Which answer gives the best measurement of the length of the metal bar shown, using the ruler shown with the bar.

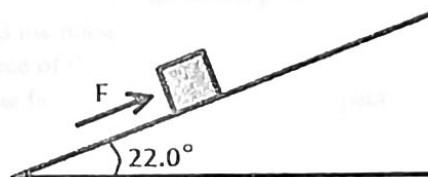


A) 1.45 cm
B) 6.43 cm
C) 6.43126 cm
D) 6.5 cm
E) 7.88 cm

- P06. A small car is moving at 30.0 mph when it gets on the entrance ramp for a highway. The car accelerates at a rate of 9500 mi/hr^2 for the entire 0.15-mile length of the ramp. What is the speed of the car as it enters the highway?

A) 36 mph
B) 48 mph
C) 53 mph
D) 61 mph
E) 73 mph

- P07. You are pushing a 50.0kg crate of zucchini up a frictionless inclined plane. The plane is angled at 22.0° with respect to the horizontal, and the 200.0N force you apply to the crate is parallel to the plane (as shown). What is the velocity of the crate after 2.50 seconds?

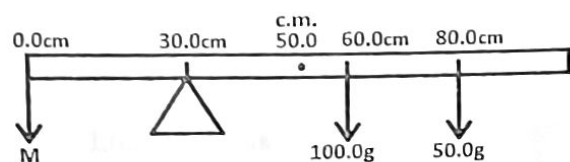


A) 0.822 m/s
B) 4.90 m/s
C) 9.18 m/s
D) 10.0 m/s
E) 14.5 m/s

- P08. A small 150.0g cart rolls frictionlessly across a horizontal table at a speed of 1.50 m/s (to the right). The cart collides with a large 220.0g cart that is stationary. After the collision, the small cart bounces directly backwards (to the left) with a speed of 0.250 m/s. What is the speed of the large cart after the collision?

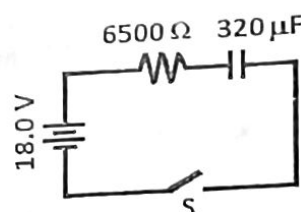
A) 0.852 m/s
B) 1.02 m/s
C) 1.19 m/s
D) 1.83 m/s
E) 2.57 m/s

- P09. A 40.0g meter stick with its center-of-mass at the 50.0cm mark is set up with a fulcrum at the 30.0cm mark (as shown). Two known masses are hung from the meter stick: a 100.0g mass at the 60.0cm mark and a 50.0g mass at the 80.0cm mark. An unknown mass is hung from the 0.0cm end, and the entire system is balanced. What is the value of the unknown hanging mass at the 0.0cm end?

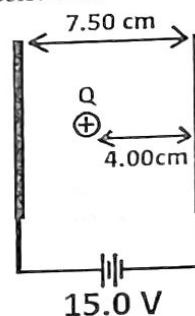


- A) 400.0g
B) 333.3g
C) 267.7g
D) 210.0g
E) 183.3g
- P10. A firecracker explosion releases 13.0W of power as pure sound. The sound pulse expands spherically as it moves away from the source. What is the intensity (in decibels) of the sound at a distance of 4.20m from the explosion?
- A) 92.8 dB
B) 97.7 dB
C) 108 dB
D) 114 dB
E) 120 dB
- P11. A large plastic disk with a volume of 0.157m^3 is placed in pure water, and a 12.0kg rock is placed on top of the disk. The disk, with the rock on top, floats on the water, with 50% of the disk showing above the surface. What is the density of the plastic disk?
- A) 424 kg/m^3
B) 500 kg/m^3
C) 576 kg/m^3
D) 618 kg/m^3
E) 924 kg/m^3

- P12. For the RC circuit shown below, the switch S is closed at time $t = 0$, and the capacitor is initially uncharged. What is the charge stored on the capacitor 3.00 seconds after the switch is closed?

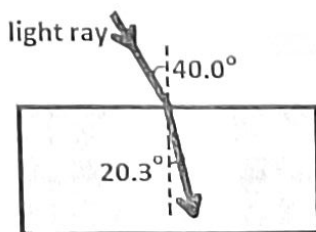


- A) 5470 μC
B) 4400 μC
C) 2880 μC
D) 1360 μC
E) 720 μC
- P13. Two parallel plates, separated by 7.50cm, are connected to a 15.0V power supply, as shown. A charged piece of dust starts at rest between the plates, at a distance of 4.00cm from the negative plate. The charge on the piece of dust is $+30.0\mu\text{C}$ and the mass of the piece of dust is 12.0mg. The piece of dust accelerates to the negative plate – how fast is it moving when it impacts the negative plate? Ignore air resistance.



- A) 20.0 m/s
B) 17.3 m/s
C) 14.6 m/s
D) 10.5 m/s
E) 6.32 m/s
- P14. A square loop of wire is placed in a 70.0 Gauss magnetic field such that the field is perpendicular to the face of the loop. The length of a side of the square loop is 18.0cm, and the resistance of the loop is 1.25Ω . Suddenly, in a time of 9.00milliseconds, the field strength increases to 430.0 Gauss. What is the magnitude of the current induced in the loop by this sudden field increase?
- A) 104 mA
B) 124 mA
C) 155 mA
D) 202 mA
E) 576 mA

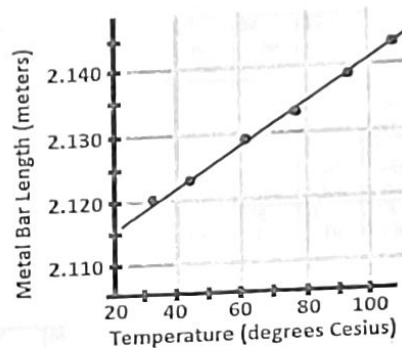
- P15. You observe the following refraction of a light ray as it passes from air into a crystal.



Based on the observed angles, what is the speed of the light in the crystal?

- A) 1.32×10^8 m/s
 B) 1.62×10^8 m/s
 C) 2.45×10^8 m/s
 D) 3.00×10^8 m/s
 E) 5.56×10^8 m/s
- P16. While visiting a modern art exhibit, you notice that a polished metal sphere in one of the sculptures acts as a convex mirror. When you stand 1.80m in front of the sphere, you observe that your image in the sphere is 1/10 of your normal height. What is the magnitude of the radius of curvature of the polished metal sphere?
- A) 16.4cm
 B) 20.0cm
 C) 32.7cm
 D) 40.0cm
 E) 80.0cm
- P17. Light with a wavelength of 460nm is directed onto the surface of a photoelectric metal. The workfunction of the metal is 1.60eV. What is the velocity of the photoelectrons emitted from the metal surface?
- A) 1.45×10^4 m/s
 B) 4.39×10^5 m/s
 C) 6.21×10^5 m/s
 D) 7.50×10^5 m/s
 E) 9.74×10^5 m/s
- P18. A particle you are seeking consists of an up quark and an anti-charm quark ($u\bar{c}$). What are the expected properties of this particle?
- A) Meson, +1 charge, -1 charm, 0 strangeness
 B) Meson, 0 charge, -1 charm, 0 strangeness
 C) Meson, -1 charge, +1 charm, -1 strangeness
 D) Baryon, +1 charge, -1 charm, 0 strangeness
 E) Baryon, 0 charge, -1 charm, +1 strangeness

- P19. You are testing the thermal properties of a new metal alloy. For the first test, you carefully measure the length of a bar of this metal alloy while heating it. A graph of the data is shown below. Based on the data, what is the approximate coefficient of thermal expansion of this metal alloy around 60°C?



- A) $1.5 \times 10^{-4} \text{ } ^\circ\text{C}^{-1}$
 B) $2.5 \times 10^{-4} \text{ } ^\circ\text{C}^{-1}$
 C) $3.3 \times 10^{-4} \text{ } ^\circ\text{C}^{-1}$
 D) $4.0 \times 10^{-4} \text{ } ^\circ\text{C}^{-1}$
 E) $6.8 \times 10^{-4} \text{ } ^\circ\text{C}^{-1}$
- P20. You are working with an optometrist and have been asked to help correct a patient's nearsightedness. Here are some data collected regarding this patient's vision:

Far Point	33.0 cm
Near Point	8.00 cm
Pressure	17 mmHg
Astigmatism	None

What power of corrective contact lens is needed to correct this patient's nearsightedness?

- A) +5.88 diopter
 B) +4.13 diopter
 C) -1.25 diopter
 D) -2.13 diopter
 E) -3.0 diopter

58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm (145)	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0
90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)

Standard Reduction Potentials

$\text{Ag}^+(\text{aq}) + \text{e}^- \rightarrow \text{Ag}(\text{s})$	$E^\circ = +0.80 \text{ V}$
$\text{Zn}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Zn}(\text{s})$	$E^\circ = -0.76 \text{ V}$

$R = 0.08206 \text{ L}\cdot\text{atm/mol}\cdot\text{K}$
 $R = 8.314 \text{ J/mol}\cdot\text{K}$
 $R = 62.36 \text{ L}\cdot\text{torr/mol}\cdot\text{K}$
 $e = 1.602 \times 10^{-19} \text{ C}$
 $N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$
 $k = 1.38 \times 10^{-23} \text{ J/K}$
 $h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$
 $c = 3.00 \times 10^8 \text{ m/s}$
 $R_H = 2.178 \times 10^{-18} \text{ J}$
 $m_e = 9.11 \times 10^{-31} \text{ kg}$

Physics

Useful Constants

quantity	symbol	value
Free-fall acceleration	g	9.80 m/s^2
Permittivity of Free Space	ϵ_0	$8.854 \times 10^{-12} \text{ C}^2/\text{Nm}^2$
Permeability of Free Space	μ_0	$4\pi \times 10^{-7} \text{ Tm/A}$
Coulomb constant	k	$8.99 \times 10^9 \text{ Nm}^2/\text{C}^2$
Speed of light in a vacuum	c	$3.00 \times 10^8 \text{ m/s}$
Fundamental charge	e	$1.602 \times 10^{-19} \text{ C}$
Planck's constant	h	$6.626 \times 10^{-34} \text{ Js}$
Electron mass	m_e	$9.11 \times 10^{-31} \text{ kg}$
Proton mass	m_p	$1.67265 \times 10^{-27} \text{ kg}$ 1.007276 amu
Neutron mass	m_n	$1.67495 \times 10^{-27} \text{ kg}$ 1.008665 amu
Atomic Mass Unit	amu	$1.66 \times 10^{-27} \text{ kg}$ $931.5 \text{ MeV}/c^2$
Gravitational constant	G	$6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$
Stefan-Boltzmann constant	σ	$5.67 \times 10^{-8} \text{ W/m}^2\text{K}^4$
Universal gas constant	R	$8.314 \text{ J/mol} \cdot \text{K}$ $0.082057 \text{ L} \cdot \text{atm/mol} \cdot \text{K}$
Boltzmann's constant	k_B	$1.38 \times 10^{-23} \text{ J/K}$
Speed of Sound (at 20°C)	v	343 m/s
Avogadro's number	N_A	$6.022 \times 10^{23} \text{ atoms/mol}$
Electron Volts	eV	$1.602 \times 10^{-19} \text{ J/eV}$
Distance Conversion	miles $\rightarrow$ meters	$1.00 \text{ mile} = 1609 \text{ meters}$
Rydberg Constant	R_∞	$1.097 \times 10^7 \text{ m}^{-1}$
Standard Atmospheric Pressure	1 atm	$1.013 \times 10^5 \text{ Pa}$
Density of Pure Water	ρ_{water}	1000.0 kg/m^3